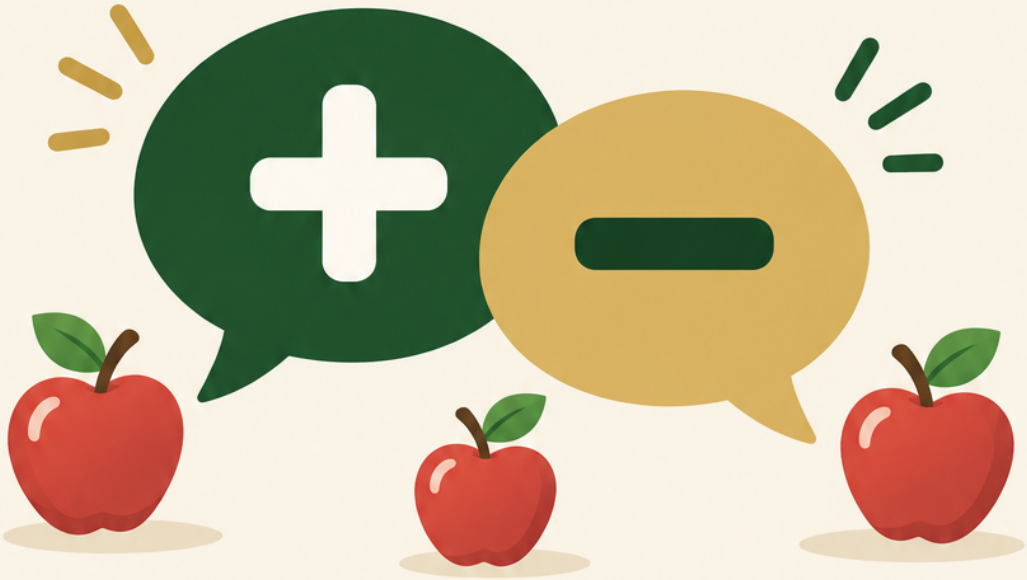
SCHOOL OF MATH

LITTLE NEWTONS • CLASS 19

Word Problems with Addition and Subtraction

Which Operation and Why?



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1. Understanding the Story

Read. Picture. Find the question.

Mia has 14 stickers. She gives 6 stickers to Leo. How many stickers does Mia have left?

Think: Will the answer be greater or less than 14?

1

What do we know?

Mia starts with 14 stickers and gives away 6.

2

What happens?

Some stickers leave Mia's group.

3

What do we need to find?

How many stickers Mia has left.

Understand the story before choosing an operation.

Understanding the story

Q1. Ben has 12 apples. He eats 3 of them. What do we know, what happens, and what do we need to find? Then solve.

Q2. Sofia has 9 shells. She finds 5 more shells at the beach. Will the answer be greater or less than 9? Why? Then solve.

Q3. There are 15 birds on a fence. Then 7 birds fly away. Will the answer be greater or less than 15? Find how many birds are still on the fence.

Q4. Read the story: “Tia has 11 pencils. Her dad gives her 6 more pencils. How many pencils does Tia have now?” Which sentence is the question? What do we need to find?

Q5. Max says: “Jada has 18 grapes. She eats 4 grapes. So she has 22 grapes.” Without solving, how can you tell Max is wrong? What is the correct answer?

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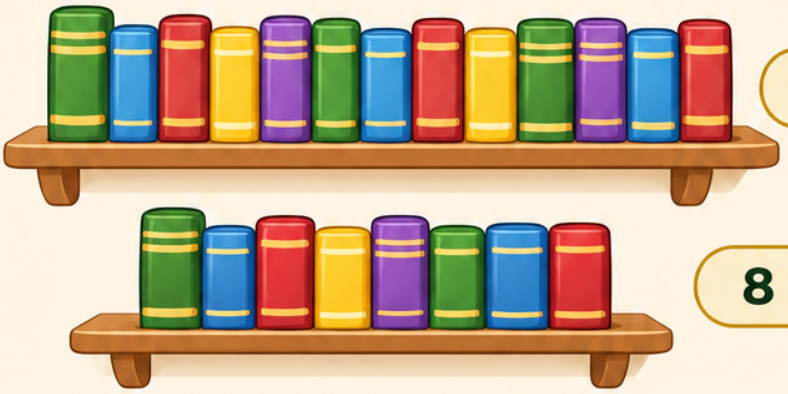
2. Putting Groups Together

Two parts make one whole.

There are 13 books on one shelf and 8 books on another shelf. How many books are on the two shelves altogether?



13 books

8 books

1 Find the parts

13 books and 8 books.

2 Choose the operation

Add because we are joining two groups.

3 Find the whole

13 + 8 = 21 books.

Make a ten: 13 + 7 + 1 = 20 + 1 = 21

Addition combines parts to find the total.

Putting groups together

Q6. There are 9 red cars and 6 blue cars in a parking lot. How many cars are there altogether?

Q7. A class has 14 girls and 7 boys. How many children are in the class in all?

Q8. A farmer has 8 cows and 12 sheep. How many animals does the farmer have in all?

Q9. One bag has 17 oranges. Another bag has 5 oranges. Name the two parts and the whole.

Q10. Use make a ten to solve 18 + 6. Show each step.

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3. Adding More to a Group

Start with a group. Add more. Find the new total.

There are 16 children on the playground. Then 7 more children arrive. How many children are on the playground now?

1

Find the start

16 children are already there.

2

Notice the change

7 children join the group.

3

Choose addition

16 + 7 = 23 children.

Make a ten: 16 + 4 + 3 = 20 + 3 = 23

We add because the group gets bigger.

Adding more to a group

Q11. There are 11 ducks in a pond. Then 5 more ducks swim over. How many ducks are in the pond now?

Q12. Kai has 15 stickers. His friend gives him 8 more stickers. How many stickers does Kai have now?

Q13. A bus has 19 people on it. At the next stop, 4 more people get on. How many people are on the bus now?

Q14. “Ella read 13 pages in the morning. In the evening, she read 9 more pages.” Find the start, the change, and the total number of pages she read.

Q15. There were 27 balloons at a party. Then 6 more balloons were blown up. Lee writes $27 - 6$. Is he right? Explain and solve.

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4. Taking Away from a Group

Start with a group. Take some away. Find what remains.

Mia has 23 crayons. She gives 8 crayons to Leo. How many crayons does Mia have left?

1

Find the start

Mia starts with 23 crayons.

2

Notice the change

8 crayons leave her group.

3

Choose subtraction

23 − 8 = 15 crayons.

Subtract to ten: 23 − 3 − 5 = 20 − 5 = 15

We subtract because some crayons are taken away.

Taking away from a group

Q16. There are 13 cookies on a plate. Sam eats 5 cookies. How many cookies are left?

Q17. There are 21 children on the playground. Then 6 children go home. How many children are still on the playground?

Q18. Nora has 24 beads. She uses 7 beads to make a bracelet. How many beads does she have left?

Q19. A branch has 30 leaves. Then 9 leaves fall off. How many leaves are still on the branch?

Q20. Leo has 22 toy cars. He gives 8 cars to his cousin. Explain why this story uses subtraction, then solve.

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5. Comparing Two Groups

Find how many more or how many fewer.

Mia has 24 stickers. Leo has 17 stickers. How many more stickers does Mia have than Leo?

1 Compare the groups

Mia has 24.
Leo has 17.

2 Choose subtraction

$24 - 17 = 7$

3 Explain the difference

Mia has 7 more stickers than Leo.

| Count up: 17 + 3 = 20; 20 + 4 = 24; 3 + 4 = 7

Leo has 7 fewer stickers than Mia.

Subtract to find the difference between two groups.

Comparing two groups

Q21. Ava has 15 stickers. Ben has 9 stickers. How many more stickers does Ava have than Ben?

Q22. A red tower has 12 blocks. A blue tower has 20 blocks. How many fewer blocks does the red tower have?

Q23. Jon has 26 cards. Maya has 19 cards. How many more cards does Jon have than Maya?

Q24. Box A has 18 crayons. Box B has 25 crayons. How many fewer crayons are in Box A?

Q25. Zoe has 14 shells and Tim has 21 shells. Tim says: “The question says how many more, so I add: 14 + 21 = 35.” What is wrong? How many more shells does Tim have?

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6. Finding a Missing Part

Know the whole? Find the part you do not know.

A box holds 25 crayons. 17 crayons are blue. The rest are red. How many crayons are red?

1 Find the whole

25 crayons in all.

2 Find the known part

17 blue crayons.

3 Find the missing part

25 – 17 = 8 red crayons.

Check with addition: 17 + 8 = 25

Whole – known part = missing part

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Finding a missing part

Q26. There are 18 balloons. 10 balloons are yellow, and the rest are green. How many balloons are green?

Q27. A fish tank has 22 fish. 15 fish are orange, and the rest are striped. How many fish are striped?

Q28. A bag holds 30 marbles. 12 marbles are blue, and the rest are red. How many marbles are red?

Q29. Ana had 16 grapes. She ate some grapes. Now she has 9 grapes. How many grapes did she eat?

Q30. Fill in the blank: 14 + ___ = 23. Then write the subtraction that finds the missing part.

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7. Choosing the Operation

Explain what the numbers mean.

Story A: A group grows

Leo has 18 marbles. He gets 7 more. How many marbles does he have now?

18 + 7 = 25

I add because 7 marbles join his group.

Story B: Compare two groups

Mia has 25 marbles. Leo has 18 marbles. How many more marbles does Mia have?

25 − 18 = 7

I subtract to find the difference between the groups.

Both stories use “more.” The story tells us which operation to choose.

I choose ___ because ___.

Choosing the operation

Q31. “Liam has 13 toy cars. He gets 9 more cars.” Do you add or subtract? Finish the sentence “I choose ___ because ___,” then solve.

Q32. “Liam has 22 toy cars. Noah has 13 toy cars. How many more cars does Liam have?” Do you add or subtract? Explain, then solve.

Q33. “There are 25 children in the pool. Then 8 children get out.” Finish “I choose ___ because ___,” then find how many children are still in the pool.

Q34. “A box has 17 crayons. Grace puts 8 more crayons in the box.” Choose the operation, explain why, and solve.

Q35. Which story needs subtraction? A: 14 frogs sit on a log, and 6 more frogs hop on. B: 14 frogs sit on a log, and 6 frogs hop off. C: 14 green frogs and 6 brown frogs sit on a log; how many frogs are there in all? Solve the subtraction story.

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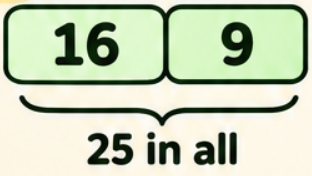
8. Solving Mixed Word Problems

Model. Choose. Solve. Check.

A. Put groups together

A class reads 16 books on Monday and 9 books on Tuesday. How many books do they read in all?

Model:



Equation: 16 + 9 = 25 books

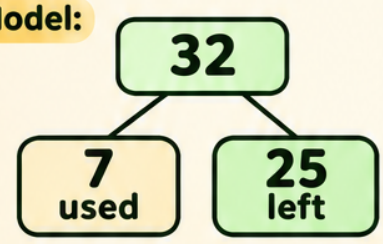
Reason: Add to combine the two groups.

Check: 25 is greater than either part.

B. Find what remains

A basket has 32 apples. 7 apples are used for a snack. How many apples remain?

Model:



Equation: 32 – 7 = 25 apples

Reason: Subtract because some apples are used.

Check: 25 + 7 = 32.

Your turn: Mia has 21 cards. Leo has 14. How many more cards does Mia have?

Read the story. Choose an operation. Explain why.

Solving mixed word problems

Q36. A library cart has 34 books. A helper puts 8 books on the shelves. How many books are still on the cart? Model, choose, solve, and check.

Q37. A class planted 17 flowers on Monday and 8 flowers on Tuesday. How many flowers did they plant in all? Model, choose, solve, and check.

Q38. Ruby has 31 beads. Kim has 24 beads. How many fewer beads does Kim have than Ruby?

Q39. A jar had some cookies. After 9 cookies were eaten, 15 cookies were left. How many cookies were in the jar at the start?

Q40. Dev has 19 stamps. He buys 6 more stamps, then gives 5 stamps to his sister. How many stamps does Dev have now?

Homework review

R1. Oscar has 17 blocks. He gives 5 blocks to his brother. Will the answer be greater or less than 17? Then find how many blocks Oscar has left.

R2. One basket has 8 apples. Another basket has 7 apples. How many apples are there altogether? Use make a ten.

R3. There are 14 bees on a flower. Then 9 more bees fly over. How many bees are on the flower now?

R4. A plate has 25 strawberries. The children eat 7 strawberries. How many strawberries are left?

R5. Luis has 23 shells. Pia has 16 shells. How many more shells does Luis have than Pia?

Homework review

R6. A drawer has 20 socks. 13 socks are white, and the rest are black. How many socks are black? Check with addition.

R7. Ivy has 16 ribbons. Max has 9 more ribbons than Ivy. How many ribbons does Max have? Finish “I choose ___ because ___.”

R8. A box has 33 pencils. The class uses 6 pencils. Model, choose, solve, and check: how many pencils are left?

Homework word problems

W1. A pet shop has 7 cats and 8 dogs. How many pets does it have in all?

W2. Zara picked 16 flowers. She gave 4 flowers to her teacher. How many flowers does Zara have now?

W3. A train has 18 passengers. At the station, 7 more passengers get on. How many passengers are on the train now?

W4. Owen has 27 crayons. Ella has 20 crayons. How many fewer crayons does Ella have than Owen?

W5. There are 24 children in the gym. 15 children are jumping rope, and the rest are playing ball. How many children are playing ball?

Homework word problems

W6. A baker made 32 muffins. She sold 8 muffins. How many muffins are left?

W7. Kim had some marbles. She found 6 more marbles. Now she has 21 marbles. How many marbles did Kim have at first?

W8. Ty has 13 stickers. Ada has 8 more stickers than Ty. How many stickers does Ada have?

W9. A garden has 28 tulips. Rabbits eat 9 tulips. How many tulips remain? Check your answer.

W10. A bus has 15 children. At the first stop, 8 children get on. At the next stop, 6 children get off. How many children are on the bus now?

Homework word problems

W11. Sam has 12 red blocks and 9 blue blocks. He uses 7 blocks to build a tower. How many blocks are not in the tower?

W12. Challenge: Mia has 18 stickers. Leo has 5 fewer stickers than Mia. How many stickers do Mia and Leo have together?